

Supplementary Material for The Perceptual Cost of Passthrough: How Video See-Through HMDs Degrade Human Visual Perception of Acuity, Contrast, and Color

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Repository: <https://github.com/Chaosikaros/VST-Visual-Perception-Benchmark>

Supplementary Statistical Tables

This supplementary material provides the detailed statistical tables moved from the main manuscript for readability. Q3, QP, VP, and eyes denote Quest 3, Quest Pro, Vision Pro, and naked-eye viewing, respectively. P-values are Bonferroni-adjusted where noted; highlighted cells indicate significant comparisons. For the Wilcoxon post-hoc table, W denotes the Wilcoxon signed-rank statistic, and $r = |z|/\sqrt{n}$ is computed from the normal approximation using nonzero paired differences.

Table S1: Complete Friedman tests for the four viewing conditions: Quest 3, Quest Pro, Vision Pro, and naked-eye viewing.

Light Level	Metric	χ^2	df	p	Kendall's W
Normal	logMAR	34.6	3	<0.001	0.786
	logCS	32.5	3	<0.001	0.739
	TES	22.0	3	<0.001	0.500
Low	logMAR	36.0	3	<0.001	0.818
	logCS	36.0	3	<0.001	0.818
	TES	26.6	3	<0.001	0.604

Table S2: Complete Wilcoxon signed-rank post-hoc tests with Bonferroni correction and effect-size r for pairwise viewing-condition comparisons.

Metric	Pair	Normal light			Low light		
		W	r	p	W	r	p
logMAR	Q3 vs. QP	0	0.88	0.003**	0	0.88	0.003**
	Q3 vs. VP	3	0.79	0.059	0	0.88	0.003**
	Q3 vs. eyes	0	0.88	0.003**	0	0.88	0.003**
	QP vs. VP	0	0.88	0.003**	0	0.88	0.003**
	QP vs. eyes	0	0.88	0.003**	0	0.88	0.003**
	VP vs. eyes	0	0.88	0.003**	0	0.88	0.003**
logCS	Q3 vs. QP	0	0.88	0.003**	0	0.88	0.003**
	Q3 vs. VP	32	0.16	1.000	0	0.88	0.003**
	Q3 vs. eyes	0	0.88	0.003**	0	0.88	0.003**
	QP vs. VP	0	0.88	0.003**	0	0.88	0.003**
	QP vs. eyes	0	0.88	0.003**	0	0.88	0.003**
	VP vs. eyes	0	0.88	0.003**	0	0.88	0.003**
TES	Q3 vs. QP	4.5	0.78	0.023*	0	0.89	0.006**
	Q3 vs. VP	24	0.34	1.000	35.5	0.08	1.000
	Q3 vs. eyes	6.5	0.74	0.044*	4	0.79	0.018*
	QP vs. VP	2	0.83	0.018*	0	0.88	0.003**
	QP vs. eyes	0	0.88	0.003**	0	0.88	0.003**
	VP vs. eyes	5.5	0.74	0.064	3	0.79	0.059

Table S3: Complete Mann-Whitney U tests comparing normal-light and low-light groups for each metric and viewing condition.

Metric	Condition	U	p
logMAR	Quest 3	0	<0.001
	Quest Pro	0	<0.001
	Vision Pro	7	<0.001
	eyes	77	0.795
logCS	Quest 3	144	<0.001
	Quest Pro	142	<0.001
	Vision Pro	14.5	<0.001
	eyes	94.5	0.195
TES	Quest 3	101	0.10
	Quest Pro	68	0.84
	Vision Pro	88	0.37
	eyes	99.5	0.117